



The Instruction Tax on Factual Recall

Base Models Can Beat Their Instruction-Tuned Counterparts

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1 AKBC Shared Task

Closed-book knowledge base construction

Predict all valid objects for a subject–relation pair.

- **Closed-book:** no retrieval, no external data, no fine-tuning; under 32B parameters.
- **Six relations:** 477 / 475 / 475 train / val / test.
- **Nearly 20% unanswerable:** abstention is scored.

Instruction tax

Instruction tuning decreases factual recall. The cause is the **weights**, not the prompt.

hasArea probe, gemma-3-27b:
base **0.880**, tuned **0.590**.

2 Model Selection

Completion prompt

Base model. No chat template, no instruction.

The total area of Lake Victoria is 68800 square kilometres.
The total area of Bodensee is 536 square kilometres.
The total area of Vygozero is ■

One exemplar per quintile of hasArea values. Set relations:
about one third of exemplars are “none”.

ONE FORMAT PER RELATION

The seating capacity of {s} is {v}.

Countries sharing a land border with {s}: {v}

{s} stock exchange listings: {v}

Recipients of {s}: {v}

Knowledge probe

FAMILY	BASE, COMPL.	IT, CHAT	IT, COMPL.	TAX Δ	CEILING
gemma-3-12b	.720	.370	.320	+.350	.84 → .47
gemma-3-27b	.880	.590	.580	+.290	.95 → .69
gemma-2-27b	.720	.640	.640	+.080	.82 → .77
Mistral-Nemo-2407	.710	.630	.660	+.080	.90 → .87
Mistral-Small-3.2-24B	–	.720	.690	–	–
Magistral-Small-2506	–	.610	.570	–	–
Qwen3-30B-A3B (MoE)	–	.430	.430	–	–
Qwen3-30B-A3B-Think.	–	.430	.410	–	–

Voted accuracy, 100 hasArea validation rows. Δ = base – chat. Ceiling: correct in ≥ 1 sample.

Same prompt, gemma-3-27b-it: **0.580**. The tuned model forgot facts.

3 Methodology

STEP 1

Probe first

~100 known facts,
several phrasings,
8–12 samples each.

~1 GPU-hour per
model version.

STEP 2

Never ask a question

Training exemplars,
then an unfinished
line. No chat template.

+0.116 F1 over chat,
same weights.

STEP 3

Keep what repeats

64–256 samples; keep
answers above a fitted
threshold.

0.6218 (n = 8) →
0.6604 (n = 64).

STEP 4

Verify rare candidates

awardWonBy: verify
low-support
candidates (AUC 0.86).

+0.027 on
awardWonBy.

STEP 5

Dead or alive?

All 16 samples say
alive → answer empty.

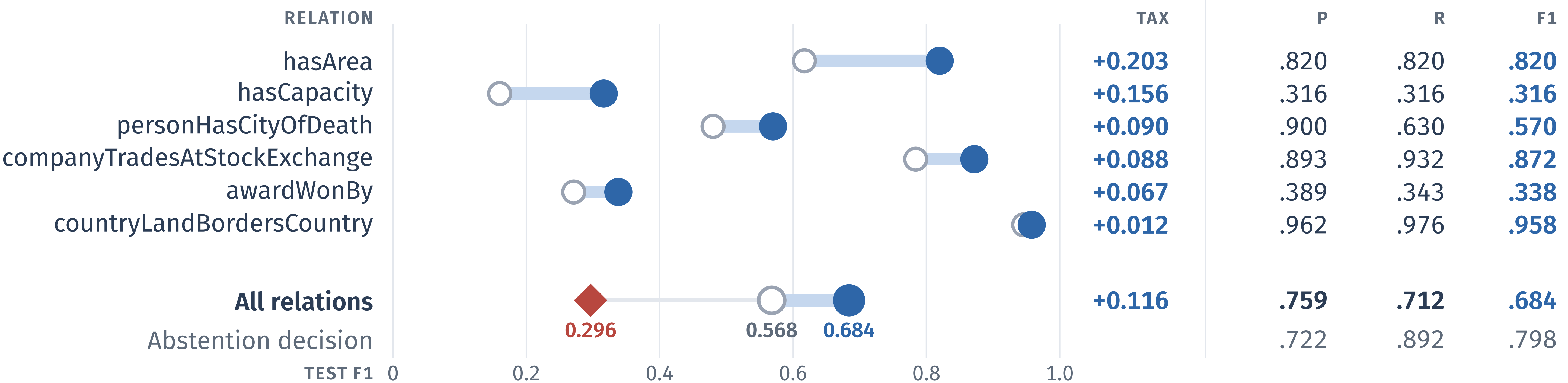
+0.030 on death;
abstention F1 0.773 →
0.798.

4 Results

Official test split · gemma-3-27b-pt (27.43B) · vLLM on two H100s

Instruction tax at full-system scale

◆ Task baseline ○ Instruction-tuned, chat ● Base, completion



TAKEAWAYS

- 1 **Tuning can make models forget facts.** Sample ceiling .95 → .69.
- 2 **Know when not to answer.** With many empty rows, it helps almost as much as knowing more.

LIMITATIONS

- Four base/instruct pairs only
- Tax size varies by tuning run
- 64–256 generations per row
- Not zero-shot
- Steps 4–5: one relation each



PAPER

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Code github.com/aiamourad/lm-kbc-2026